X-CAVATE

GUI User Guide

Version 2.0.0

Convert vascular network geometries into collision-free
3D printer toolhead pathways and G-code.

SM = Single-Material | **MM** = Multimaterial

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1 Getting Started

X-CAVATE provides a web-based GUI built with Streamlit. It requires no coding experience to operate. This section covers installation and launching the interface.

1.1 Installation

1.1.1 Option A: Conda (Recommended)

1. Open a terminal (Mac: Terminal.app; Windows: Anaconda Prompt or PowerShell).
2. Confirm conda is installed:

   ```
   conda --version
   ```

   If not installed, download Miniconda from https://docs.conda.io/en/latest/miniconda.html.
3. Clone the repository and enter the directory:

   ```
   git clone https://github.com/sohams-MASS/X-CAVATE.git
   cd X-CAVATE
   ```

4. Create and activate the conda environment:

   ```
   conda env create -f environment.yml
   conda activate xcavate-gui
   ```

5. Verify the installation:

   ```
   xcavate --help
   ```

1.1.2 Option B: pip

1. Confirm Python 3.9+ is installed: `python3 --version`
2. Clone the repository (same as above).
3. Install with GUI support:

   ```
   pip install ".[gui]"
   ```

   For CLI-only usage (no web interface): `pip install .`

1.1.3 Option C: Docker

1. Install Docker Desktop from https://www.docker.com/products/docker-desktop.

2. Build the image:

```
docker build -t xcavate .
```

3. Run the container:

```
docker run -p 8501:8501 xcavate
```

4. Open http://localhost:8501 in your browser.

1.2 Launching the GUI

After installation (conda or pip), start the server:

```
conda activate xcavate-gui      # if using conda
streamlit run xcavate/gui/app.py
```

A browser window opens automatically at **http://localhost:8501**. If it does not open, navigate to that URL manually.

Tip

To stop the server, press `Ctrl+C` in the terminal.

1.3 Dependencies

Package	Version	Purpose
numpy	≥ 1.21	Array operations
pandas	≥ 1.3	Data handling
scipy	≥ 1.7	KD-tree spatial indexing, interpolation
plotly	≥ 5.0	Interactive 3D visualization
matplotlib	≥ 3.5	Static plots
streamlit	≥ 1.20	Web GUI (optional with pip; included in conda env)

2 GUI Overview

The X-CAVATE GUI uses a two-panel layout:

Sidebar (left panel)
File upload widgets and all configurable parameters in expandable sections.

Main area (right panel)
Top-level tabs containing the pipeline controls, results, custom G-code editor, print instructions, and calibration validation.

2.1 Abbreviations

Throughout the interface:

- **SM** = Single-Material
- **MM** = Multimaterial

2.2 Tab Structure

The main area contains up to four tabs (the second appears only when the Custom G-code toggle is enabled):

Tab	Description
Pipeline	Run button, 7-step progress bar, 3D visualizations, download buttons
Custom G-code	Tabbed template editor for header, extrusion, pressure, and dwell G-code snippets (visible only when enabled)
Print Instructions	Step-by-step positioning and printing guidance for SM and MM modes
Calibration	Compute volumetric flow rate (Q) from calibration filaments and validate measured diameters against targets

3 Sidebar: Input Files

The top of the sidebar contains two file uploaders:

Widget	Accepted type	Description
Network file	`.txt`	Vascular network coordinates exported from SimVascular. Each vessel has a header line followed by coordinate rows (x, y, z in cm, plus optional radius and arterial/venous flag columns).
Inlet / Outlet file	`.txt`	Text file identifying inlet and outlet node coordinates. Each section starts with the keyword `inlet` or `outlet`, followed by (x, y, z) triplets.

Warning

Both files must be uploaded before the Run X-CAVATE button becomes active. Coordinates in the network file must be in **centimeters** — X-CAVATE converts to millimeters internally.

3.1 Network File Format

```
Vessel: 0, Number of Points: 100
33.4878, 0.0000, 53.5303, 0.0200
33.4115, -0.5285, 52.7311, 0.0200
...
Vessel: 1, Number of Points: 100
15.0600, -37.3300, 11.6600, 0.0190
...
```

Column	Required	Description
1–3	Yes	x, y, z coordinates (cm)
4	No	Vessel radius (needed for speed calculation)
5	No	Arterial/venous flag: 0 = venous, nonzero = arterial (needed for multi-material)

3.2 Inlet / Outlet File Format

```
inlet
21.97, -15.43, 50.00
11.81, 0.00, 21.34
outlet
36.05, -27.19, 50.00
50.00, 0.00, 26.54
```

Coordinates are matched to the nearest network point automatically.

4 Sidebar: Required Parameters

These parameters appear directly below the file uploaders and must be set for every run.

Parameter	Type	Default	Description
Nozzle diameter (mm)	float	0.5	Outer diameter of the print nozzle.
Container height (mm)	float	10.0	Height of the print container.
Decimal places	int	3	Number of decimal places for output coordinate rounding.
Amount up (mm)	float	10.0	Distance to raise the nozzle above the container between passes.
Printer type	select	Pressure	Hardware target: Pressure (pneumatic extrusion), Positive Ink (displacement pump), or Aerotech (6-axis motion controller).
Multimaterial	toggle	Off	Enable two-nozzle arterial/venous printing. Requires a 5-column input file (x, y, z, radius, arterial/venous flag).

5 Sidebar: Optional Parameters

Optional parameters are organized into seven expandable sections. Click a section header to reveal or hide its contents.

5.1 Tolerancing

Parameter	Type	Default	Description
Enable tolerance	toggle	Off	When on, nodes within the tolerance distance are allowed in the same pass even if one geometrically blocks the other.
Tolerance	float	0.0	Tolerance distance (mm). 0 means none.

5.2 Print Settings

Parameter	Type	Default	Description
Print speed (mm/s)	float	1.0	Nozzle travel speed during extrusion. Typical range: 0.5–5.0 mm/s.
Flow (mm^3/s)	float	0.161	Volumetric flow rate for speed calculation mode.
Jog speed (mm/s)	float	5.0	Speed for rapid non-printing travel moves between passes.
Jog speed lift (mm/s)	float	0.25	Speed for the vertical lift at the end of each pass.
Initial lift (mm)	float	0.5	Height the nozzle lifts after finishing a pass before traveling.
Dwell start (s)	float	0.08	Pause after starting extrusion before nozzle movement. *Aerotech only.*
Dwell end (s)	float	0.08	Pause after nozzle stops before extrusion off. *Aerotech only.*

Info

The **Dwell start** and **Dwell end** fields only appear when the printer type is set to **Aerotech**.

5.3 Geometry

Parameter	Type	Default	Description
Container X (mm)	float	50.0	Width of the print container. Used for centering instructions.
Container Y (mm)	float	50.0	Depth of the print container. Used for centering instructions.
Scale factor	float	1.0	Multiplier applied to all coordinates after unit conversion.
Top padding (mm)	float	0.0	Extra clearance above the highest network point for safe nozzle retraction.

5.4 Downsampling

Parameter	Type	Default	Description
Enable downsampling	toggle	Off	Reduce point density to speed up printing. Branchpoints and endpoints are preserved.
Downsample factor	int	1	Factor by which to downsample the network.

5.5 Multimaterial

These settings are relevant when the **Multimaterial** toggle is enabled.

Parameter	Type	Default	Description
Offset X (mm)	float	103.0	Horizontal distance between the two printheads.
Offset Y (mm)	float	0.5	Vertical fine-tuning offset between printheads.
Front nozzle	int	1	1 = venous nozzle in front, 2 = venous nozzle behind.
Printhead 1 name	text	Aa	Axis designation for the arterial printhead in G-code.
Printhead 2 name	text	Ab	Axis designation for the venous printhead in G-code.
Axis 1 name	text	A	Z-axis letter for the arterial printhead.
Axis 2 name	text	B	Z-axis letter for the venous printhead.
Jog translation (mm/s)	float	10.0	Speed for travel moves between nozzles during printhead switching.
Resting pressure (psi)	float	0.0	Pressure applied to the inactive printhead.
Active pressure (psi)	float	5.0	Pressure applied to the active printhead during extrusion.

5.6 Positive Ink Displacement

Parameter	Type	Default	Description
Use radii	toggle	Off	Use per-node vessel radii instead of a fixed diameter.
Diameter (mm)	float	1.0	Fixed vessel diameter for extrusion calculations.
Syringe diameter (mm)	float	1.0	Inner diameter of the syringe barrel.
Factor	float	1.0	Scaling multiplier for computed extrusion amounts.
Separate arterial/venous	toggle	Off	Set different start/end displacements per material.

When **Separate arterial/venous** is **off**, two fields appear:

Parameter	Type	Default	Description
Start (mm)	float	0.0	Plunger displacement at the start of each pass to prime ink flow.
End (mm)	float	0.0	Plunger displacement at the end of each pass (negative retracts).

When **Separate arterial/venous** is **on**, four fields appear instead:

Parameter	Type	Default	Description
Start (arterial)	float	0.0	Plunger displacement for the arterial syringe at pass start.
Start (venous)	float	0.0	Plunger displacement for the venous syringe at pass start.
End (arterial)	float	0.0	Plunger displacement for the arterial syringe at pass end.
End (venous)	float	0.0	Plunger displacement for the venous syringe at pass end.

5.7 Advanced

Parameter	Type	Default	Description
Custom G-code	toggle	Off	Enable to provide custom G-code templates (reveals the Custom G-code tab).
Pathfinding algorithm	select	DFS	DFS: thorough collision detection via depth-first search. Sweep Line: faster bottom-up approach for large networks.
Overlap nodes	int	0	Number of nodes to retrace at pass boundaries to bridge small gaps.
Overlap algorithm	select	Retrace	"Retrace" scans all previous passes (original). "Consecutive" checks adjacent pairs (faster).
Generate plots	toggle	On	Create interactive 3D HTML visualizations.
Speed calculation	toggle	Off	Compute per-node print speed from vessel radius and flow rate.
Gap closure (SM)	toggle	Off	Load gap extension files for SM prints. When on, shows file uploaders for pass indices and deltas.
Gap closure (MM)	toggle	Off	Load gap extension files for MM prints. When on, shows file uploaders for pass indices and deltas.

6 Pipeline Tab

The **Pipeline** tab is the primary workspace where you run the X-CAVATE pipeline and view results.

6.1 Running the Pipeline

1. Upload both input files in the sidebar.
2. Configure required and optional parameters.
3. Click Run X-CAVATE (full-width primary button at the top of the tab).

Info

The run button is disabled until both input files are uploaded.

6.2 Progress Bar

The pipeline executes in 7 steps. A progress bar tracks each step:

Step	Description
1	Reading network and inlet/outlet files
2	Interpolating network to nozzle resolution
3	Building adjacency graph and detecting branchpoints
4	Generating collision-free print passes
5	Post-processing: subdivision, gap closure, overlap
6	Processing multimaterial passes
7	Writing output files and G-code

Upon completion, a success message appears: "X-CAVATE finished successfully."

6.3 3D Visualizations

If **Generate plots** is enabled, three interactive 3D Plotly charts are displayed below the progress bar:

1. **Original Network** — The raw vascular network before processing, with each vessel shown as a separate trace.
2. **Single Material Print Passes** — The computed collision-free SM print paths, color-coded by pass index.
3. **Multimaterial Print Passes** — (If multimaterial is enabled) MM paths color-coded by material type (arterial vs. venous).

All plots are interactive: rotate by clicking and dragging, zoom with the scroll wheel, and hover over traces for coordinate information.

6.4 Download Results

Below the visualizations, three download columns appear:

Column	Contents
G-code	Download buttons for each generated G-code file (e.g., `gcode_SM_pressure.txt`, `gcode_MM_pressure.txt`).
Coordinate Files	Download buttons for SM/MM coordinate files (x, y, z, combined).
Changelog	Download button for `changelog.txt` (gap closure operations log).

Tip

Results persist in the browser session. You can switch to other tabs and return to download files later without re-running the pipeline.

7 Custom G-code Tab

This tab appears only when the **Custom G-code** toggle is enabled in the sidebar's Advanced section. It provides a tabbed editor for entering printer-specific G-code snippets that replace the default commands in the final output file.

7.1 Sub-tabs

The Custom G-code tab contains four sub-tabs:

Sub-tab	What to paste
Header	Printer initialization code: pressure box connections, homing commands, etc.
Single Material	Start/stop extrusion code for single-nozzle prints (2 fields, side-by-side).
Multimaterial	Start/stop extrusion + active/resting pressure for each printhead (8 fields in a 2-column layout).
Dwell	Optional time-delay code for "spot welding" at pass junctions (2 fields: start-of-pass and end-of-pass).

7.2 Template Fields

The table below maps every editable text area to the output template file it replaces:

GUI Label	Output Filename	Sub-tab
Header code	`header_code.txt`	Header
Start extrusion	`start_extrusion_code.txt`	Single Material
Stop extrusion	`stop_extrusion_code.txt`	Single Material
Start extrusion — Printhead 1	`start_extrusion_code_printhead1.txt`	Multimaterial
Start extrusion — Printhead 2	`start_extrusion_code_printhead2.txt`	Multimaterial
Stop extrusion — Printhead 1	`stop_extrusion_code_printhead1.txt`	Multimaterial
Stop extrusion — Printhead 2	`stop_extrusion_code_printhead2.txt`	Multimaterial
Active pressure — Printhead 1	`active_pressure_printhead1.txt`	Multimaterial
Active pressure — Printhead 2	`active_pressure_printhead2.txt`	Multimaterial
Resting pressure — Printhead 1	`rest_pressure_printhead1.txt`	Multimaterial
Resting pressure — Printhead 2	`rest_pressure_printhead2.txt`	Multimaterial
Dwell code (start of pass)	`dwell_start.txt`	Dwell
Dwell code (end of pass)	`dwell_end.txt`	Dwell

Tip

Snippets are saved automatically in the browser session and persist across page re-runs, even when the Custom G-code toggle is temporarily turned off.

8 Print Instructions Tab

After a successful pipeline run, the **Print Instructions** tab shows positioning and printing guidance specific to your network and parameters.

8.1 Start Position Metrics

Three metric cards display the starting coordinates:

Metric	Description
Start X (mm)	X-coordinate where the nozzle should begin.
Start Y (mm)	Y-coordinate where the nozzle should begin.
Start Z (mm)	Z-coordinate where the nozzle should begin.

8.2 Padding Metrics

Four metric cards show the container padding — how far the network is offset from each container wall:

Metric	Description
Left padding	Distance (mm) from the network to the left container wall.
Right padding	Distance (mm) from the network to the right container wall.
Front padding	Distance (mm) from the network to the front container wall.
Back padding	Distance (mm) from the network to the back container wall.

8.3 Printing Instructions

Depending on the mode, one of the following expandable sections appears:

Single Material Instructions
: Step-by-step instructions for positioning the single printhead and starting the print job.

Multimaterial Calibration
: Instructions for calibrating the offset between the two printheads.

Multimaterial Positioning
: Instructions for positioning both printheads and starting the multimaterial print job.

Info

This tab shows "Run the pipeline first to see print instructions" until you have completed a successful pipeline run.

9 Calibration Tab

The **Calibration** tab provides two features: computing the volumetric flow rate (Q) from calibration measurements and validating measured filament diameters against target values.

9.1 Compute Volumetric Flow Rate (Q)

This section calculates the average volumetric flow rate from calibration filament measurements and produces a calibration curve (radius vs. speed). During calibration, pressure is held constant while the print speed is varied for each target diameter group.

9.1.1 Inputs

Widget	Type	Description
Upload measured diameters CSV	file upload	CSV file with a `Length` column containing measured filament diameters in **microns**.
Print speeds used during calibration	text input	Comma-separated list of print speeds in mm/s, one per target diameter group (same order). Pressure is held constant; speed is varied for each calibration filament.
Target diameters	text input	Comma-separated list of expected filament diameters in **mm**, in the same order as the measurements.
Measurements per target	int (default: 3)	Number of rows in the CSV to average for each target diameter.

9.1.2 Running Q Computation

Click Compute Q to process. The button is disabled until a CSV file is uploaded.

9.1.3 Outputs

Upon completion, the section displays:

1. The computed Q value as a metric (mm^3/s).
2. An interactive Plotly chart showing the calibration curve (radius vs. speed).
3. Download buttons for:
   - **Q_pressure.txt** — the computed Q value.
   - **radii_pressure.html** — interactive calibration curve.

9.2 Calibration Validation

This section lets you compare measured filament diameters against target values to verify print accuracy.

9.2.1 Inputs

Widget	Type	Description
Upload measured diameters CSV	file upload	CSV file with a `Length` column containing measured filament diameters in **microns**.
Target diameters	text input	Comma-separated list of expected filament diameters in **mm**, in the same order as the measurements.
Measurements per target	int (default: 3)	Number of rows in the CSV to average for each target diameter.

9.2.2 Running Validation

Click Run Validation to process. The button is disabled until a CSV file is uploaded.

9.2.3 Outputs

Upon completion, the section displays:

1. An interactive Plotly chart comparing target vs. measured diameters.
2. Download buttons for:
   - **CSV report** (`calibration_error.csv`) — tabular error analysis.
   - **HTML report** (`calibration_error.html`) — interactive visualization.

10 Output Files

All output is written to a temporary directory. Use the download buttons in the **Pipeline** tab to save files locally. When running via CLI, files are written to the `outputs/` directory (or the path specified by `--output_dir`).

10.1 Directory Structure

```
outputs/
  graph/
    graph.txt                # Adjacency graph
    special_nodes.txt        # Branchpoints, endpoints, inlets, outlets
    SM_x_coords.txt          # Single-material x coordinates per pass
    SM_y_coords.txt          # Single-material y coordinates per pass
    SM_z_coords.txt          # Single-material z coordinates per pass
    SM_combined.txt          # Combined xyz per pass
    MM_x_coords.txt          # Multimaterial x coordinates (if enabled)
    MM_y_coords.txt          # Multimaterial y coordinates (if enabled)
    MM_z_coords.txt          # Multimaterial z coordinates (if enabled)
    MM_combined.txt          # Multimaterial combined xyz (if enabled)
  gcode/
    gcode_SM_pressure.txt    # Single-material G-code
    gcode_MM_pressure.txt    # Multimaterial G-code (if enabled)
  plots/
    network_original.html    # 3D plot of original network
    network_SM.html          # 3D plot of SM print passes
    network_MM.html          # 3D plot of MM print passes (if enabled)
  changelog.txt              # Gap closure operations log
```

Info

The G-code filename suffix matches your selected printer type: `pressure`, `positive_ink`, or `aerotech`.

10.2 File Descriptions

File Description	Directory
`graph.txt` Full adjacency graph of the interpolated network.	`graph/`
`special_nodes.txt` Identified branchpoints, endpoints, inlets, and outlets.	`graph/`
`SM_x_coords.txt` X-coordinates for each SM print pass, one pass per line.	`graph/`
`SM_y_coords.txt` Y-coordinates for each SM print pass.	`graph/`
`SM_z_coords.txt` Z-coordinates for each SM print pass.	`graph/`
`SM_combined.txt` Combined (x, y, z) coordinates per SM pass.	`graph/`
`MM_*.txt` Multimaterial equivalents of the above (if enabled).	`graph/`
`gcode_SM_*.txt` Single-material G-code for the selected printer type.	`gcode/`
`gcode_MM_*.txt` Multimaterial G-code (if enabled).	`gcode/`
`network_original.html` Interactive 3D visualization of the original vascular network.	`plots/`
`network_SM.html` Interactive 3D visualization of SM print passes.	`plots/`
`network_MM.html`	`plots/`

File Description	Directory
Interactive 3D visualization of MM print passes.	
`changelog.txt` Log of all gap closure operations performed.	root

11 Troubleshooting

Problem Solution
`Runtime instance already exists` (GUI) A stale Streamlit process is running. Kill it with `pkill -f streamlit`, then restart the server.
Inlet/outlet matching warnings Inlet/outlet coordinates are approximate. X-CAVATE matches to the nearest network point and logs the distance. Verify the matches are reasonable.
Pipeline hangs on large networks Reduce coordinates per vessel to 100 or fewer, or enable downsampling. The Sweep Line algorithm may also be faster for very large networks.
`ModuleNotFoundError` Run `pip install ".[gui]"` to install all dependencies, or recreate the conda environment.
Docker container won't start

Problem **Solution**
Ensure Docker Desktop is running and port 8501 is not in use by another process.
Download buttons disappear after clicking Update to the latest version of X-CAVATE (this has been fixed).
No G-code files generated Verify that both input files are correctly formatted and that the pipeline completed without errors. Check the terminal for error messages.
Plots not showing Ensure **Generate plots** is enabled in the Advanced section. If disabled, plots are skipped for faster execution.
Browser does not open Navigate manually to `http://localhost:8501`. On remote servers, ensure port 8501 is forwarded.